**EXP NO.:** 3
**DATE:**

# CISCO PACKET TRACER

## AIM:

To analyse the behaviour of network devices using **CISCO PACKET TRACER SIMULATOR**.

## PROCEDURE:

1. From the network component box, click and drag the following components: **4 generic PCs and one switch**.
2. To make new connections:
    - **a.** Click on the **Copper Straight-Through cable** .
    - **b.** Select a port on the PC and connect it to the switch using the cable. Then the **Link LED** should glow in green.
    - **c.** Similarly, link the other three computers.
3. Click on the connected PC, go to the **Desktop** tab, click on **IP Configuration** , and enter an **IP address and subnet mask**.
4. Click on the **PDU (message icon)** and send it to the PCs.
5. Observe the flow of the PDU from source PC to destination PC by selecting the **Real-Time/Simulation mode** .

## HOW TO DOWNLOAD AND INSTALL CISCO PACKET TRACER :

Cisco provides Packet Tracer through its learning platforms. Cisco's installation instructions state that Packet Tracer is available for **Windows, Linux, and macOS**.

### Step 1: Create/Login to an Account

1. Go to Cisco Skills for All.
2. Create a free account or log in with your existing Cisco/Skills for All account.
3. You can also access Packet Tracer through Cisco Networking Academy when provided by your course.

Cisco documentation notes that a **NetAcad.com or SkillsForAll.com account may be required to authenticate when launching Packet Tracer** .

### Step 2: Download Packet Tracer

1. Open the **Resources** section.
2. Select **Download Packet Tracer**.
3. Choose the installer appropriate for your operating system:
    - **Windows**
    - **macOS**
    - **Linux**
4. Save the installation file to your computer.

Cisco's official instructions provide installers for these desktop operating systems.

### Step 3: Install on Windows

1. Open the folder where the downloaded installer is saved.

2. Double-click the Packet Tracer installation file.
3. Read and accept the **License Agreement**.
4. Click **Next**.
5. Select the installation location or keep the default location.
6. Select the Start Menu folder.
7. Choose whether to create additional shortcuts.
8. Click **Install**.
9. Wait for the installation to finish.
10. Click **Finish**.
11. Open **Cisco Packet Tracer** from the Start Menu or desktop shortcut.

These steps follow Cisco's published Windows installation procedure.

**Step 4: Install on Ubuntu/Linux**

1. Download the appropriate **.deb** Packet Tracer installer.
2. Open **Terminal**.
3. Go to the folder containing the downloaded file.
4. Run:

```
sudo apt-get install ./CiscoPacketTracer_xxx_Ubuntu_64bit.deb
```

Replace `xxx` with the version number of the downloaded Packet Tracer file.

1. Follow the License Agreement prompts.
2. Accept the agreement.
3. Complete the installation.
4. Launch Packet Tracer from the applications menu.

Cisco's installation instructions specifically provide the `apt-get install ./...deb` method for Ubuntu.

**Step 5: Install on macOS**

1. Download the macOS version of Packet Tracer.
2. Open the downloaded installer.
3. Follow the installation prompts.
4. Accept the Cisco License Agreement.
5. Complete the installation.
6. Open **Cisco Packet Tracer** from Applications.
7. Sign in using your Cisco/NetAcad/Skills for All account when prompted.

**Step 6: Start the Experiment**

After installation:

1. Open **Cisco Packet Tracer** .
2. Select **End Devices**.
3. Drag **4 PCs** into the workspace.
4. Select **Switches** and drag **one switch** into the workspace.
5. Select **Connections**.
6. Choose the **Copper Straight-Through** cable.
7. Connect each PC to the switch.
8. Assign IP addresses, for example:
    - PC0: 192.168.1.1
    - PC1: 192.168.1.2
    - PC2: 192.168.1.3
    - PC3: 192.168.1.4
    - Subnet mask: 255.255.255.0
9. Use **Simulation Mode** to observe how packets travel between the computers.

## RESULT:

Thus, the behaviour of network devices was analysed and studied with **CISCO PACKET TRACER SIMULATOR** .